IIQ to JPEG Converter

Installation and User Guide

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This guide provides step-by-step instructions for installing and using the IIQ to JPEG batch converter tool.

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1. Overview

The IIQ to JPEG Converter is a batch processing tool designed to convert Phase One IIQ RAW image files to high-quality JPEG format. The tool automatically matches the brightness and color characteristics of the camera's original preview, ensuring output that is identical to what you see in IX Capture or other Phase One software.

Key Features:

- Batch conversion of thousands of IIQ files
- Parallel processing for maximum speed (up to 64 threads)
- Automatic brightness matching using camera preview
- Full EXIF metadata preservation (camera settings, GPS, date/time)
- Resume capability - automatically continues interrupted conversions
- Folder structure preservation option
- Detailed batch logging
- No manual Python installation required

2. System Requirements

Component	Requirement
Operating System	Windows 7 or later (64-bit)
Processor	Multi-core CPU recommended
RAM	8 GB minimum, 16 GB+ recommended
Disk Space	~150 MB for software + space for output files
Internet	Required for first-time setup only (~50 MB download)

Note: The tool runs entirely from a local folder and does not require administrator privileges or system-wide Python installation.

3. Installation Instructions

3.1 First-Time Setup

The installation is automatic and takes approximately 2-3 minutes on first run.

Step 1: Obtain the Files

You will receive two files from your administrator:

- IIQ_Converter_Encrypted.bat - The launcher
- iiq_converter_encrypted.pye - The application

Step 2: Create a Folder

Create a folder on your computer where you want the tool installed. For example:

```
C:\IIQ_Converter\
```

Step 3: Place Files

Copy both files into a folder, example below:

```
C:\IIQ_Converter\
├── IIQ_Converter_Encrypted.bat
└── iiq_converter_encrypted.pye
```

Step 4: Run First-Time Setup

Double-click IIQ_Converter_Encrypted.bat

A command window will appear showing installation progress:

- Downloading Python 3.11.9 (~12 MB)
- Installing required packages
- This takes about 2-3 minutes

Some Python program folders will be created inside the folder.

> **Note:** The _python_env folder contains the Python environment. Do not delete it or the tool will not run.

Step 5: First Launch

After installation completes, the converter window will open automatically. You're ready to start converting images!

4. Using the Converter

4.1 Starting the Application

After the first-time setup, launching the converter is instant:

- Double-click IIQ_Converter_Encrypted.bat
- The converter window opens immediately (no setup needed)

4.2 Converting Images

1. Select Input Folder

Click Browse next to 'Input Folder' and navigate to the folder containing your IIQ files. The tool will search all subfolders automatically.

2. Select Output Folder

Click Browse next to 'Output Folder' and choose where you want the converted JPEG files saved.

3. Adjust Settings (Optional)

You can adjust quality, threads, and other options (see section 4.3 below). The default settings work well for most cases.

4. Start Conversion

Click ▶ Start Conversion

The progress area will show:

- Real-time log of each file being processed
- Progress bar
- Success/failure counts

5. Wait for Completion

Processing speed depends on:

- Number of files
- Image size
- CPU cores available
- Number of parallel threads selected

A popup appears when conversion finishes.

6. Review Output

Navigate to your output folder to find the converted JPEG files. A log file is also saved at:

```
[Output Folder]\bin\log.txt
```

4.3 Settings Explained

Setting	Description	Recommended
JPEG Quality	Compression quality (1-100). Higher = larger files, better quality.	100 (maximum quality)
Parallel Threads	Number of images processed simultaneously. More threads = faster, but uses more CPU.	Your CPU cores - 1 (Default: auto-detected)
Preserve folder structure	Keep the input folder organization in output, or flatten all files to one folder.	Checked (preserve structure)
Colour Space	Adobe RGB has wider color gamut, sRGB is standard for web/display.	Adobe RGB (1998)

Note: Start with default settings. If conversion is slow, reduce the number of parallel threads. If output quality isn't high enough, increase JPEG quality (though 100 is already maximum).

5. Features

5.1 Automatic Resume Capability

If conversion is interrupted (computer restart, disk full, power loss), the tool automatically resumes where it left off:

- Reads the log file to identify completed images
- Skips already-processed files
- Re-processes the last incomplete batch to verify file integrity
- Continues with remaining images

Simply select the same input and output folders, then click Start Conversion again. The tool will handle the rest.

5.2 EXIF Metadata Preservation

All important metadata from your IIQ files is automatically copied to the output JPEG files:

- Camera Information: Make, model, serial number
- Capture Settings: ISO, shutter speed, aperture, focal length, date/time
- Lens Details: Lens make, model, serial number
- GPS Coordinates: Latitude, longitude, altitude (if present in IIQ)
- Copyright: Artist, copyright, creator information

This means your converted JPEGs retain all the important information needed for archival, asset management, and professional workflows.

5.3 Batch Logging

Every conversion session is logged to:

```
[Output Folder]\bin\log.txt
```

The log contains:

- Timestamp for each batch completion
- List of successfully converted files (✓)
- List of failed files with error messages (✗)
- Batch-by-batch results (one batch = number of parallel threads)

6. Troubleshooting

Problem: 'Failed to download Python' error during setup

Solution: Check your internet connection. If behind a firewall, contact your IT administrator. The tool needs to download ~50 MB on first run.

Problem: Conversion is very slow

Solution: Reduce the number of parallel threads. High thread counts can overwhelm slower CPUs. Try starting at 4 threads and increasing gradually.

Problem: Some images failed to convert

Solution: Check the log file for specific error messages. Common causes include corrupted IIQ files or insufficient disk space.

Problem: Output images are not showing EXIF data

Solution: The source IIQ files may not have had EXIF embedded. Verify using ExifTool or Windows file properties on the original IIQ.

Problem: Application won't start

Solution: Delete the _python_env folder and run the BAT file again. This will reinstall everything fresh.

Problem: 'Decompression bomb' warning

Solution: This is normal for large Phase One images (100MP+). The warning is harmless and conversion will proceed normally.

7. Uninstalling

To completely remove the IIQ to JPEG Converter:

- Close the application if it's running
- Delete the entire folder (e.g., C:\IIQ_Converter\)
- Done! The tool leaves no registry entries or system-wide changes.

Note: Uninstalling does not affect any JPEG files you've already converted. Only the application itself is removed.

8. Support

For questions, issues, or feedback about the IIQ to JPEG Converter:

Contact Person:	Bipul Neupane, PhD
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Organization:	DPIRD Node, APPN
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When reporting issues, please include:

- A description of the problem
- Any error messages shown
- The contents of bin\log.txt (if available)
- Number of files being processed
- Your computer specifications (Windows version, RAM, CPU)

This tool was developed using claude.ai

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